The convenient unit

4a

Speed up your calculations by using a calculator.

Physical measurements such as 100 km/h have two parts; a number (the size or measure, e.g. 100) and a unit (e.g. km/h). Today the units most scientists use are from the metric system.

1 Complete the table below by writing the correct unit beside each quantity, and then writing the abbreviation of each unit. The first one has been done for you.
Choose from these units: ampere, byte, joule, litre, kilogram, metre, newton, second, volt.

Quantity	Base unit name	Base unit abbreviation
Distance	*metre*	*m*
Time		
Mass		
Volume		
Weight		
Energy		
Electric current		
Voltage (potential difference)		
Computer memory size		

In order to keep the number a meaningful and friendly size, prefixes are often used as part of the unit. The 'magic prefix numbers' in the metric system are 1000 (10^3) and 0.001 (10^{-3}). Kilo is the prefix indicating a multiplier of 1000. Milli is the prefix meaning 1/1000.

1000 millilitres make up 1 litre; and a millilitre equals 1/1000 litre or 0.001 litre.

2 **a** Complete the table which shows the value of a prefix and its abbreviation.

Name	Value: number	Value: exponent	Abbreviation
milli		10^{-3}	
micro	1/1 000 000		
nano		10^{-9}	
pico	1/1 000 000 000 000		p
kilo		10^{3}	
mega	1 000 000		M
giga		10^{9}	

b Prefix abbreviations are listed in the first column of the table. Complete the table below. Provide examples using these prefixes in relation to the metre. The first one has been done for you.

Prefix	Example: name	Example with units abbreviation	Fraction of a metre	Decimal of a metre	Magnitude in index form	Number to make one metre
m	millimetre	1 mm	1/1000 m	0.001 m	10^{-3}	1000 mm
µ						
n						
k						
M						
G						

Note that centimetres (1 cm = 0.01 m; 100 cm = 1 m) and decimetres (1 dm = 0.1 m; 10 dm = 1 m) are used in some situations; however, metres and millimetres are preferred by scientists, architects, builders and trades people.

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3 People choose to use a prefix so that the number part of the measurement is convenient and meaningful. Saying that a plane trip is 3 000 000 m long is awkward. Saying 3 000 km or 3 Mm makes the expression of the distance easier to understand. Choose a better unit in which to express the measurements listed in the following table.

Distance	Measurement	Better unit
Length of a football field	100 m	
Length of a marathon running track	40 000 m	
Diameter of a tiny dot of ink	0.001 m	
Diameter of a pencil dot	0.000 1 m	
Distance from earth to moon	300 000 000 m	
Distance between earth and sun	150 000 000 000 m	
Thickness of an onion cell	0.000 05 m	
Length of a virus	0.000 000 2 m	

4 **a** Measuring time is a little more complicated. It is common to use prefixes with seconds (s) – but only when the time is less than a second. For larger time measurements we switch to minutes (min) and hours (h) and then to days (days) and years (y). Complete the table below about time measurements.

Unit name	Unit symbol	Equivalent in seconds	Number per second
millisecond			1 000 ms
microsecond		1/1000 000	
minute	min		
hour			
day			
year			

b Express the time measurements for each event in the table in a more meaningful unit.

Event	Time	Standard measurement
Light travelling from sun to earth	480 s	
Sound travelling from Sydney to Canberra	1 020 s	
Car driving from Melbourne to Canberra	21 600 s	
Student doing serious science homework	1 800 s	
Driver having a micro sleep	0.001 s	
Electric energy moving from power point to television	0.000 000 01 s	
Rock falling off a cliff	0.002 h	
Time for a dog to eat its dinner	0.01 h	

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